

STAT 341: Tutorial Test 1 - Sample 0

First name

Last name

Signature

ID Number

Email

Time allowed: 50 minutes

Special instructions

- All numeric answers should be written either as a number to three decimal places or as a fraction reduced to its lowest terms.
- If working is not specifically requested, a correct answer will receive full marks. However, an incorrect answer could still receive credit if working is provided. An incorrect answer with no working will receive zero marks.

1. (6 marks) The population standard deviation is $\sigma_y = \sqrt{\sum_{u \in P} (y_u - \bar{y})^2 / N}$

(a) (2 marks) Show that the population standard deviation is location invariant.

For $P+a = \{y_1+a, \dots, y_N+a\}$ where $a \in \mathbb{R}$,

$$\bar{y}_{P+a} = \frac{\sum_{u \in P} (y_u + a)}{N} = a + \bar{y}.$$

$$\text{Hence, } \sigma_y(P+a) = \sqrt{\frac{\sum_{u \in P} (y_u + a - (a + \bar{y}))^2}{N}} = \sqrt{\frac{\sum_{u \in P} (y_u - \bar{y})^2}{N}} = \sigma_y. //$$

(b) (2 marks) Show that the population standard deviation is scale equivariant.

For $cP = \{cy_1, \dots, cy_N\}$ where $c > 0$,

$$\bar{y}_{cP} = \frac{\sum_{u \in P} cy_u}{N} = c\bar{y}.$$

$$\text{Hence, } \sigma_y(cP) = \sqrt{\frac{\sum_{u \in P} (cy_u - c\bar{y})^2}{N}} = \sqrt{c^2} \sqrt{\frac{\sum_{u \in P} (y_u - \bar{y})^2}{N}} = c\sigma_y. //$$

(c) (1 mark) Show that population coefficient of variation, which is the population standard deviation divided by the average, is scale invariant.

From part b, we know that $\bar{y}_{cP} = c\bar{y}$ and $\sigma_y(cP) = c\sigma_y$.

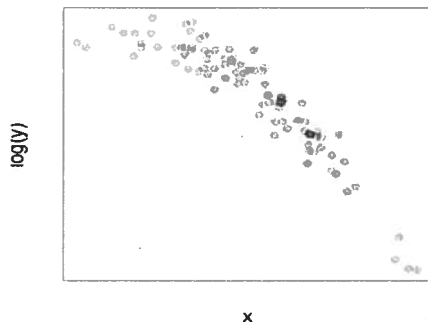
$$\text{Hence, } \frac{\bar{y}_{cP}}{\sigma_y(cP)} = \frac{c\bar{y}}{c\sigma_y} = \frac{\bar{y}}{\sigma_y}. //$$

(d) (1 mark) Give an advantage of the coefficient of variation over the population standard deviation.

Coefficient of variation is scale-free, as shown in part c). //

2. (5 marks) Power transformations.

(a) (2 marks) Consider the scatterplot below of $\log y$ versus $\log x$:



Which of the following transformations might straighten this scatterplot?

Circle those that do apply.

Cross out those that do not.

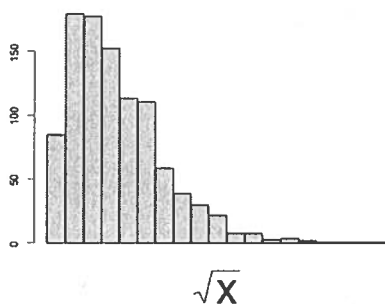
(i) $\log(y)$ vs $\log(x)$

(ii) y^2 vs $\log(x)$

(iii) $\log(y)$ vs x^2

(iv) y vs $\sqrt{x}$

(b) (2 marks) Consider the histogram below of $\log x$:



Which of the following transformations might make this histogram more symmetric?

Circle those that do apply.

Cross out those that do not.

(i) $\log(x)$

(ii) $-\frac{1}{x}$

(iii) x^2

(iv) x

(c) (1 mark) Why do power transformations preserve the ranks of the population?

Power transformations are monotonic for all α ,
the ranks of the population does not change
after transformation. //

3. (5 marks) Consider the location statistic:

$$T_N(y_1, \dots, y_N) = \bar{y} = \frac{1}{N} \sum_{i=1}^N y_i$$

(a) (3 marks) Derive this statistic's sensitivity curve:

$$SC(y) = N(T_N(y_1, \dots, y_{N-1}, y) - T_{N-1}(y_1, \dots, y_{N-1}))$$

$$\text{Given } T_{N-1}(y_1, \dots, y_{N-1}) = \frac{1}{N-1} \sum_{i=1}^{N-1} y_i = \bar{y}_{N-1},$$

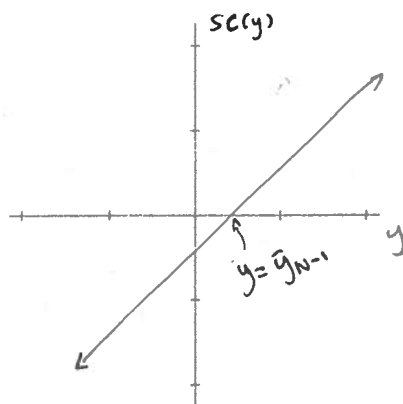
$$T_N(y_1, \dots, y_{N-1}, y) = \frac{1}{N} \left(\sum_{i=1}^{N-1} y_i + y \right)$$

$$= \frac{N-1}{N} \bar{y}_{N-1} + \frac{y}{N}$$

$$\therefore SC(y) = N \left(\frac{N-1}{N} \bar{y}_{N-1} + \frac{y}{N} - \bar{y}_{N-1} \right)$$

$$= y - \bar{y}_{N-1}$$

(b) (1 mark) Give a sketch of the sensitivity curve: (c) (1 mark) From this curve, what do you conclude about this statistic's resistance to outliers in y ?



It's not resistant against outliers,
since the curve is unbounded. //

4 marks

4. (4 marks) Implicitly defined attributes.

(a) (1 mark) Give an example of an implicitly defined population attribute.

Least-squares parameter θ : $\operatorname{argmin}_{\theta \in \mathbb{R}} \sum_{u \in \mathcal{P}} (y_u - \theta)^2$, or

solution to $\sum_{u \in \mathcal{P}} \psi(y_u; \theta) = 0$, where $\psi = \frac{\partial}{\partial \theta} \rho$. //

(b) (3 marks) In point form, describe the gradient descent algorithm.

Objective: minimize $\rho(\theta)$ wrt. θ

Initialize: $i=0$, initial estimate $\hat{\theta}_0$.

Loop for iteration i :

1. Compute gradient: $g_i = \frac{\partial}{\partial \theta} \rho(\theta) \big|_{\theta = \hat{\theta}_i}$

2. Gradient direction: $g_i = g_i / \|g_i\|$

3. Line search: $\hat{\lambda}$ such that $\hat{\lambda} = \operatorname{argmin}_{\lambda} \rho(\hat{\theta} - \lambda g_i)$

4. Update parameter: $\hat{\theta}_{i+1} = \hat{\theta}_i - \hat{\lambda} g_i$

5. Test for convergence:

if converged, STOP LOOP

else: $i = i+1$, REPEAT LOOP

Output: $\hat{\theta} = \hat{\theta}_i$ //

5. [5 marks] Determine whether the following statements are true or false.

(a) A scatterplot is not an attribute.

i. True

ii. False

(b) The least squares line is replication invariant.

i. True

ii. False

(c) A histogram is an approximation of a quantile plot

i. True

ii. False

(d) It is preferred for a spread attribute to be location-scale invariant.

i. True

ii. False

(e) A skewness attribute (a measure of asymmetry) should be both scale and location invariant.

i. True

ii. False